

ANDREW N. SIVAPRAKASAM

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[Website](#) | [GitHub](#) | [Twitter](#)

EDUCATION

Indiana University School of Medicine
MD/PhD Candidate
Medical Scientist Training Program (MSTP)

August 2018-Present
Indianapolis, IN

Purdue University
PhD in Biomedical Engineering

April 2020-March 2025
West Lafayette, IN

Thesis: *Place and Time Processing of Pitch in the Context of Cochlear Dysfunction*

University of Pittsburgh
B.S.E. in Bioengineering
Minors: Computer Science, Chemistry
Magna Cum Laude

August 2013-April 2018
Pittsburgh, PA

EXPERIENCE

Auditory Neurophysiology and Modeling & SNAP Laboratories May 2019 - March 2025
PhD Student— Advisors: Michael Heinz, PhD & Hari Bharadwaj, PhD *West Lafayette, IN*

- I utilized my background in engineering, signal processing, and music to study the processing of pitch by the auditory system in listeners with and without sensorineural hearing loss (SNHL). By leveraging controlled animal models of SNHL, I interpreted patterns in behavior and electrophysiology relevant to the neural processing of pitch and periodicity in human listeners.

Through this work, I identified specific perceptual consequences that could be linked to deficits in cochlear tonotopic organization (place) and temporal coding (time) deficits.

EMG-Based Aid for Dysphagia Rehabilitation September 2021 - May 2022
Software Consultant for Georgia Malandraki, PhD *West Lafayette, IN*

- Refined a novel device to improve impaired deglutition after stroke or injury. Provided programming (C#) expertise to translate swallowing EMG signal recordings into an intuitive, front-end patient interface.

Brainstem Neurophysiology Laboratory July 2018 - August 2018
Rotation Student — Advisor: Mark Sayles, MD, PhD *West Lafayette, IN*

- Observed methods used to record single-unit auditory nerve spikes at the Purdue Dept. of Speech, Language, & Hearing Sciences.

Human Engineering Research Laboratories (HERL) January 2016 - April 2018
Research Associate — Advisor: Alicia Koontz, PhD *Pittsburgh, PA*

- Assisted with study design, data analysis, and programming related to investigating wheelchair propulsion kinetics and kinematics using the Computer-Assisted Rehabilitation Environment (CAREN). Led SBIR Phase I development and testing of the *AgileLife Patient Transfer System*.

Orthopaedic Biodynamics Laboratory May 2014 - December 2015
Research Assistant — Advisor: Scott Tashman, PhD *Pittsburgh, PA*

- Studied knee joint biomechanics and cartilage morphology after ACL reconstruction using Dynamic Stereo X-ray, MRI, and CT image analysis techniques

GRANTS & FELLOWSHIPS

National Institutes of Health

- *Place and Time Processing of Pitch in the Context of Cochlear Dysfunction*.
Ruth L. Kirschstein Predoctoral MD/PhD Fellowship (1F30DC020916). 01/01/2023-12/31/2025.
Role: Principal Investigator. Total Direct Costs: \$158,082

Purdue University

- Interdisciplinary Training Program in Auditory Neuroscience (1T32DC016853) Fellowship (2020-2022)

Indiana University

- Haselby Family Scholarship Recipient (2018-2020)

University of Pittsburgh

- Swanson School of Engineering Summer Research Internship 2017
- Swanson School of Engineering Summer Research Internship 2015

AWARDS & ACHIEVEMENTS

Academic

- 2023 ARO MidWinter Meeting Travel Award Recipient
- Finalist for University of Pittsburgh *Co-op Student of the Year*
- Nominated for the 2018 George Washington Prize at University of Pittsburgh

Other

- 2023 Honorable Mention in Color Photography, Professional Division, Indiana State Fair

TEACHING EXPERIENCE

Purdue University

- Guest Lecturer & Teaching Assistant, BME 511 — Biomedical Signal Processing
Lecture Topics: Linear algebraic approaches, Eigendecomposition, Principal Component Analysis
[Student Evaluation](#): 4.14 / 5

University of Pittsburgh

- Teaching Assistant, HRS 2774 — Rehabilitation Biomechanics (Fall 2016)
- Teaching Assistant, BIOENG 1310 — Bioinstrumentation (Spring 2016 & 2017)

PEER-REVIEWED PUBLICATIONS

1. **A. Sivaprakasam**, I. Schweinzer, M. Heinz, “Comparing Physiological Signatures of Synaptopathic Noise Exposure and Carboplatin-Induced Inner-Hair-Cell Dysfunction in Chinchillas”. *Hearing Research* (2026) DOI: [10.1016/j.heares.2026.109739](https://doi.org/10.1016/j.heares.2026.109739).
2. F. Deloche, **A. Sivaprakasam**, M. Heinz, “Characterization of cochlear compressive nonlinearities with forward-masked compound action potentials”. 19th International Symposium on Hearing. Zenodo (2024). DOI: [10.5281/zenodo.6576993](https://doi.org/10.5281/zenodo.6576993)
3. S. Hauser, **A. Sivaprakasam**, H. Bharadwaj, M. Heinz, “Precision Diagnostics for Complex Sensorineural Hearing Loss,” *Mechanics of Hearing Workshop 2024 (MoH 2024)*. DOI: [10.5281/zenodo.13334652](https://doi.org/10.5281/zenodo.13334652)
4. A. Farhadi, S. Hauser, **A. Sivaprakasam**, M. Heinz, “Simultaneous Recording of Otoacoustic Emissions and Envelope-Following Responses to Evaluate Efferent Influences on Neural Coding,” *Mechanics of Hearing Workshop 2024 (MoH 2024)*. DOI: [10.5281/zenodo.13329441](https://doi.org/10.5281/zenodo.13329441)

5. F. Deloche, S. Parida, **A. Sivaprakasam**, M. Heinz, “Estimation of Cochlear Frequency Selectivity Using a Convolution Model of Forward-Masked Compound Action Potentials,” *Journal of the Association for Research in Otolaryngology* (2024). DOI: [10.1007/s10162-023-00922-1](https://doi.org/10.1007/s10162-023-00922-1)
6. N. Patel, D. Wilson, C. Russell III, B. Johnson, **A. Sivaprakasam**, “On Empowering Medical Students to Become Innovators”. *Insight: Indiana University Medical Student Research Journal* Vol. 7 No. 1 (2025). pp 22.
7. **A. Sivaprakasam**, H. Wang, R. Cooper, and A. Koontz, “Innovation in Transfer Assist Technologies for Persons with Severe Disabilities and Their Caregivers,” *IEEE Potentials*, vol. 36, no. 1, pp. 34-41. DOI: [10.1109/MPOT.2016.2614761](https://doi.org/10.1109/MPOT.2016.2614761)

SELECTED CONFERENCE PRESENTATIONS

Perceptual and Physiological Consequences of Cochlear Hearing Loss

1. S. Hauser, J. Mondul, **A. Sivaprakasam**, H. Bharadwaj, M. Heinz “Predicting Profiles of Cochlear Pathology from Non-Invasive Physiological Biomarkers.” Association for Research in Otolaryngology 49th Annual MidWinter Meeting, San Juan, Puerto Rico, 2026.
2. S. Hauser, **A. Sivaprakasam**, H. Bharadwaj, M. Heinz “Multi-Biomarker Battery for Stratifying Complex Sensorineural Hearing Loss and Predicting Listening Outcomes.” Association for Research in Otolaryngology 49th Annual MidWinter Meeting, San Juan, Puerto Rico, 2026.
3. M. Heinz, A. Farhadi, **A. Sivaprakasam**. “Inner-hair-cell dysfunction as another form of hidden hearing loss that degrades speech-in-noise coding” Midwest Auditory Research Conference, Rootstown, OH, 2025.
4. M. Heinz, A. Farhadi, **A. Sivaprakasam**. “Inner-hair-cell dysfunction as another form of hidden hearing loss that degrades speech-in-noise perception” International Symposium on Hearing (ISH), Vienna, Austria, 2025.
5. **A. Sivaprakasam**, S. Hauser, M. Heinz, H. Bharadwaj, “Integrating Physiological and Perceptual Assays to Resolve the Effects of Sensorineural Hearing Loss on Neural Place and Time Cues for Pitch.” Association for Research in Otolaryngology 48th Annual MidWinter Meeting, Orlando, FL, 2025.
6. S. Hauser, **A. Sivaprakasam**, H. Bharadwaj, M. Heinz, “Detecting and Disambiguating ‘Hidden’ Cochlear Pathologies: Inner Hair Cell Injury and Cochlear Synaptopathy.” Association for Research in Otolaryngology 48th Annual MidWinter Meeting, Orlando, FL, 2025.
7. S. Hauser, **A. Sivaprakasam**, H. Bharadwaj, M. Heinz, “Physiological Profiling of Sensorineural Hearing Loss for Predicting Speech-in-Noise Outcomes for hearing in noise.” American Auditory Society, Scottsdale, AZ, 2025.
8. **A. Sivaprakasam**, S. Hauser, H. Bharadwaj, M. Heinz, “Subtypes of Sensorineural Hearing Loss Variably Alter the Neural Encoding of Fundamental Frequency.” Auditory System Gordon Research Conference (GRC). Smithfield, RI, 2024.
9. **A. Sivaprakasam**, H. Bharadwaj, M. Heinz, “Cross-Species Investigations of Place and Time Coding of Pitch Using Envelope-Following Responses.” Association for Research in Otolaryngology 47th Annual MidWinter Meeting, Anaheim, CA, 2024.
10. A. Farhadi, S. Hauser, **A. Sivaprakasam**, M. Heinz, “Evaluation of Efferent Influences on Neural Coding Using Pre-Clinical Models of Sensorineural Hearing Loss.” 186th Acoustical Society of America Meeting. Ottawa, ON, 2024.

11. A. Farhadi, M. Patra, **A. Sivaprakasam**, M. Heinz, “Investigating Potential Sources of Modulation Enhancement in Noise through Physiologically Recorded and Model Neural Responses.” Association for Research in Otolaryngology 47th Annual MidWinter Meeting, Anaheim, CA, 2024.
12. **A. Sivaprakasam**, I. Schweinzer, H. Bharadwaj, M. Heinz, “Upper-Harmonic Deficits in Temporal Envelope Coding of Tone Complexes and Amplitude Modulations Differentiate Inner Hair Cell Damage From Synaptopathy,” Association for Research in Otolaryngology 46th Annual MidWinter Meeting, Orlando, FL, 2023.
13. **A. Sivaprakasam**, I. Schweinzer, H. Bharadwaj, M. Heinz, “Inner Hair Cell Damage and Cochlear Synaptopathy Differentially Impact Neural Envelope Coding of Modulations and Pitch,” 9th Midwest Auditory Research Conference, Ann Arbor, MI, 2022. (Platform)
14. I. Schweinzer, **A. Sivaprakasam**, M. Heinz, “Differentiating Inner Hair Cell Dysfunction From Cochlear Synaptopathy Using Non-Invasive Measures of Temporal Envelope Coding in Chinchilla,” Association for Research in Otolaryngology 45th Annual MidWinter Meeting, San Jose, CA, 2022.

Computational Modeling of Neurophysiology & Frequency Selectivity

15. P. Darveshi, A. Farhadi, S. Hauser, **A. Sivaprakasam**, M. Heinz. “Deep Learning–Based Classification of Hidden Cochlear Pathologies based on ABR Waterfall Plots,” 7th Virtual Conference on Computational Audiology 2026. DOI: [10.5281/zenodo.20735748](https://doi.org/10.5281/zenodo.20735748)
16. P. Darveshi, A. Farhadi, S. Hauser, **A. Sivaprakasam**, M. Heinz. “Deep learning–based classification of cochlear pathologies using ABR waterfall plots in chinchilla models,” Association for Research in Otolaryngology 49th Annual MidWinter Meeting, San Juan, Puerto Rico, 2026.
17. P. Darveshi, A. Farhadi, S. Hauser, **A. Sivaprakasam**, M. Heinz. “Deep learning–based classification of cochlear pathologies using ABR waterfall plots in chinchilla models,” Midwest Auditory Research Conference, Rootstown, OH, 2025.
18. M. Patra, A. Farhadi, **A. Sivaprakasam**, D. Axe, M. Heinz, “Refinement of Inner Hair Cell Dysfunction in a Phenomenological Auditory Nerve Model Using Physiological and Single-Unit Recordings following Selective IHC Dysfunction.” Association for Research in Otolaryngology 47th Annual MidWinter Meeting, Anaheim, CA, 2024.
19. M. Patra, **A. Sivaprakasam**, D. Axe, M. Heinz, “Testing Phenomenological Auditory-Nerve Model Predictions for Selective Inner- and Outer-Hair-Cell Dysfunction.” 184th Acoustical Society of America Meeting. Chicago, IL, 2023.
20. F. Deloche, **A. Sivaprakasam**, M. Heinz, “Effect of Suppressive Masking on the Dynamic Range of Auditory-Nerve Responses: Characterization With Forward-Masked Compound Action Potentials,” Association for Research in Otolaryngology 46th Annual MidWinter Meeting, Orlando, FL, 2023.
21. F. Deloche, S. Parida, **A. Sivaprakasam**, M. Heinz, “Estimation of Cochlear Frequency Selectivity Using a Convolution Model of Forward-Masked Compound Action Potentials,” Association for Research in Otolaryngology 45th Annual MidWinter Meeting, San Jose, CA, 2022.
22. **A. Sivaprakasam**, H. Bharadwaj, M. Heinz, “The Role of Envelope and Temporal Fine Structure in Auditory Neural Coding of Timbre in Normal and Impaired Hearing,” Neuroscience 2021, Chicago, IL, 2021.

Translational Research & Large-Scale Data Analysis

23. M. Heinz, A. Grama, S. Hauser, **A. Sivaprakasam**, H. Bharadwaj, O. Clavier “An Open-Science Cross-Species Data Resource for Standardized Hearing Assessments.” Association for Research in Otolaryngology 48th Annual MidWinter Meeting, Orlando, FL, 2025.

24. A. Hustedt-Mai, J. Alexander, E. Bartlett, A. Grama, M. Shader, J. Simpson, E. Jensen, S. Hauser, **A. Sivaprakasam**, M. Heinz “Accessible Precision Audiology Research Center (APARC) in a Community-based Innovation Districts.” American Auditory Society, Scottsdale, Arizona, 2025.
25. V.M. Athreya, **A. Sivaprakasam**, H. Ginsberg, H. Bharadwaj, M. Heinz, “Pioneering Cortical Assays of Gap Detection to Explore Temporal Processing in Chinchilla using a Multi-Channel Mini-EEG Cap.” Association for Research in Otolaryngology 47th Annual MidWinter Meeting, Anaheim, CA, 2024.
26. **A. Sivaprakasam**, S. Hauser, N. Seidl, D. Hake, H. Bharadwaj, M. Heinz, “Towards an Open-Source Precision Audiological Diagnostics Core for Large-Scale Data Analysis.” Virtual Conference on Computational Audiology 2023.
27. **A. Sivaprakasam**, V.M. Athreya, H. Ginsberg, H. Bharadwaj, M. Heinz, “A Chinchilla Mini-EEG Cap Improves Cross-Species Translation for Cortical and Subcortical Evoked Potentials,” Association for Research in Otolaryngology 46th Annual MidWinter Meeting, Orlando, FL, 2023.

Medical Education & Innovation

28. C. Russell III, N. Patel, D. Wilson, B. Johnson, **A. Sivaprakasam**, C. Earl, S. Conlon, Innovative Skills Development in Medical Students Through Neonatal Intubation Solutions,” IUSM Education Day, Indianapolis, IN, 2025.
29. Z. Guckien, A. Woloshuk, N. Patel, **A. Sivaprakasam**, A. Warrick, L. Garcia, “Advancing Innovation in Medicine: Expanding the Physician’s Toolbox” IUSM Education Day, Indianapolis, IN, 2021. **Awarded Outstanding Abstract for Poster Presentation**
30. A. Woloshuk, **A. Sivaprakasam**, N. Patel, A. Warrick, A. Witten, L. Brennan, Z. Guckien, N. Diggins, L. Garcia, L. Wang, J. Acchiardo, J. Merrell, “A Prototype ECG for Neonatal Resuscitation,” IUSM Student Research Symposium, Indianapolis, IN, 2020.

Rehabilitation & Orthopaedic Biomechanics (Univ. of Pittsburgh)

31. **A. Sivaprakasam**, S. Bass, D. Kamaraj, and A. Koontz, “Investigating Wheelchair Seating Parameters and Their Effect on Ramp Propulsion,” Biomedical Engineering Society 2017 Annual Meeting, Phoenix, AZ, 2017.
32. **A. Sivaprakasam**, R. Cooper, and A. Koontz, “Evaluation of the AgileLife Patient Transfer and Movement System,” Rehabilitation Engineering and Assistive Technology Society of North America (RESNA) 2017, New Orleans, LA, 2017.
33. **A. Sivaprakasam** and A. Koontz, “The Usability of an Automated Patient Transfer Technology,” 19th Annual NPSF Congress, Orlando, Florida, 2017.
34. **A. Sivaprakasam**, J. Irrgang, F. Fu, and S. Tashman, “Contralateral Limb Differences in Knee Kinetics After Anterior Cruciate Ligament Reconstruction,” Biomedical Engineering Society 2015 Annual Meeting, Tampa, FL, 2015. (Platform)
35. **A. Sivaprakasam**, J. Irrgang, F. Fu, and S. Tashman, “Contralateral Limb Differences in Knee Kinetics and Correlations to Kinematic Differences After Anterior Cruciate Ligament Reconstruction,” University of Pittsburgh Science 2015, Pittsburgh, PA, 2015.
36. E. Thorhauer, K. Sass, **A. Sivaprakasam**, J. Irrgang, F. Fu, and S. Tashman, “Alterations in In Vivo Knee Cartilage Contact After Anterior Cruciate Ligament Reconstruction and Correlations to Clinical Outcomes,” Orthopaedic Research Society 2015 Annual Meeting, Las Vegas, NV, 2015.
37. E. Thorhauer, K. Sass, **A. Sivaprakasam**, J. Irrgang, F. Fu, and S. Tashman, “Changes in Tibiofemoral Gait Kinematics Are Associated with Regional Cartilage Morphological Changes,” Orthopaedic Research Society 2015 Annual Meeting, Las Vegas, NV, 2015.

38. R. OHara-Plotnik, E. Thorhauer, **A. Sivaprakasam**, J. Irrgang, F. Fu, and S. Tashman, “Gait Is a Poor Task Choice for Identifying Kinematic Deficits After ACL Reconstruction ” Orthopaedic Research Society 2015 Annual Meeting, Las Vegas, NV, 2015.

OTHER TALKS & WORKSHOPS

1. *Subtypes of Cochlear Impairment Differentially Impact the Neural Processing and Perception of Pitch.* **A. Sivaprakasam**. IU Otolaryngology Grand Rounds. June 17th, 2026.
2. *Envelope Coding Degradation Following IHC Damage: Data vs Model.* **A. Sivaprakasam**, M. Heinz. Auditory Nerve Modeling Session hosted by Laurel Carney at the ARO 46th Annual MidWinter Meeting. February 13th, 2023.
3. *Put the PRO in Programming.* **A. Sivaprakasam**, N. Patel, A. Petrucciani. Student-led Workshop. IU Medical Scientist Training Program Retreat 2021. July 10th, 2021.

MENTORING EXPERIENCE

Fernando Aguilera de Alba, *Behavioral Comparison of Comodulation Masking Release (CMR) Between Chinchillas and Humans*. Summer 2020 (Co-Mentored), currently PhD student in Heinz Lab.

EXTRACURRICULAR INVOLVEMENT

Purdue College of Engineering

Photographer

Fall 2022-Spring 2025

West Lafayette, IN

- Hired to capture portraits, campus events, and architecture for Purdue use and publication
- Work regularly featured on [@purdueengineers](#)

Advancing Innovation in Medicine (AIM)

Co-Founder

Spring 2019-Present

Indianapolis, IN

- Student Interest Group (SIG) created with the purpose of educating medical students on the medical design process/prototyping
- Taught basic programming, CAD, design thinking, concepts using student-led design projects

Auditory Neuroscience Association at Purdue (ANAP)

President (2021-2022) | Vice President (2020-2021)

Summer 2020-Present

West Lafayette, IN

- Group focused on peer discussion, networking, and mentoring for Purdue undergraduates, graduates, and post-doctoral trainees interested in auditory neuroscience
- Organized monthly talks, discussions, and social events

Competitive Long Distance Running

Fall 2009-Present

- Regularly compete in events ranging from 5k-Marathon.
- Average 95th Percentile in best event (Half Marathon) [[Athlinks Profile](#)]
- Developed a publicly available running analysis toolbox [[Garmin Wrapped](#)]

IUSM Combined Degree Student Council (CDSC)

Class Representative

Fall 2018-Present

Indianapolis, IN

- Convey class concerns and feedback to administration and assist with organizing annual MSTP retreat
- Curated and edited video specifically for student recruitment and virtual interviews

IUSM Symphony Orchestra

Violinist

Fall 2018-Fall 2020

Indianapolis, IN

- Performed approximately twice per semester
- Continued 15 consecutive years as an orchestral musician

LICENSURE & CERTIFICATION

USMLE Step 1 — Pass, June 2020

USMLE Step 2 — Pass, May 2026

Scores available upon request

TECHNICAL STRENGTHS

Programming Literacy	Matlab, Python, HTML, R, Java, L ^A T _E X
Adobe Creative Cloud	Lightroom, Illustrator, Photoshop
Other	RPvdsEx (for Tucker-Davis Technologies Hardware), LabView